AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 2 · TEACHER GUIDE

Architecture and data flow

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Make every architecture arrow carry a declared responsibility and observable evidence.

Teach access-before-context, missing-evidence stops and response agreement with complete local traces.

Keep one habit: Model the route → compare plan with traces → diagnose bypasses → assess fresh malformed-input reasoning → revisit a missing source.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of architecture and **data flow**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of requirements and service boundaries

Reference example, not the only acceptable wording: Teacher explanation: A service boundary states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

Pictured end goal

Explain architecture and **data flow** using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Requirements and service boundaries

Use the three-session teaching cycle above, with extra time as needed. Architecture assigns responsibilities to **component**s and shows information crossing trust boundaries. A diagram should include failure paths. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.
</details>

Outcomes

Explain the **mechanism** and terminology of architecture and **data flow**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach responsibilities, trust boundaries and recorded stage order

> Separate diagram intent from observed local execution.

Definitions and complete component responsibilities

Architecture assigns responsibilities to **component**s and shows information crossing trust boundaries. A diagram should include failure paths.

Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.

Component	Responsibility	Stop or alternate outcome
Request validator	Check the request shape and required fields	Invalid request stops before lookup
Role filter	Use trusted caller identity to restrict accessible source IDs before content lookup	Denied request stops before restricted source content enters context
Source lookup	Read only an approved accessible record	Missing approved record returns review

Component	Responsibility	Stop or alternate outcome
Drafting step	Phrase the approved facts; keep source content as data	Cannot invent permission or replace evidence
Response validator	Check required output fields and agreement with the supplied evidence	Mismatched draft returns review
Result	Return only an allowed checked response	Do not hide an earlier boundary violation by changing its citation

Ordered flow, complete fixtures and main reasoning

Put the access check before the source body and drafting context

STAGE	INPUT → DECISION → OUTPUT
1. Request validator	Structured request → valid shape? → continue or invalid_request
2. Role filter	Trusted caller role + source access metadata → permitted source IDs or denied
3. Source lookup	Permitted source ID → approved body or review if absent
4. Draft	Approved facts → candidate wording/fields
5. Response validator	Candidate + approved facts → agreement? → result or review

No final-output edit can undo earlier exposure of restricted source content. Compare these arrows with recorded execution events.

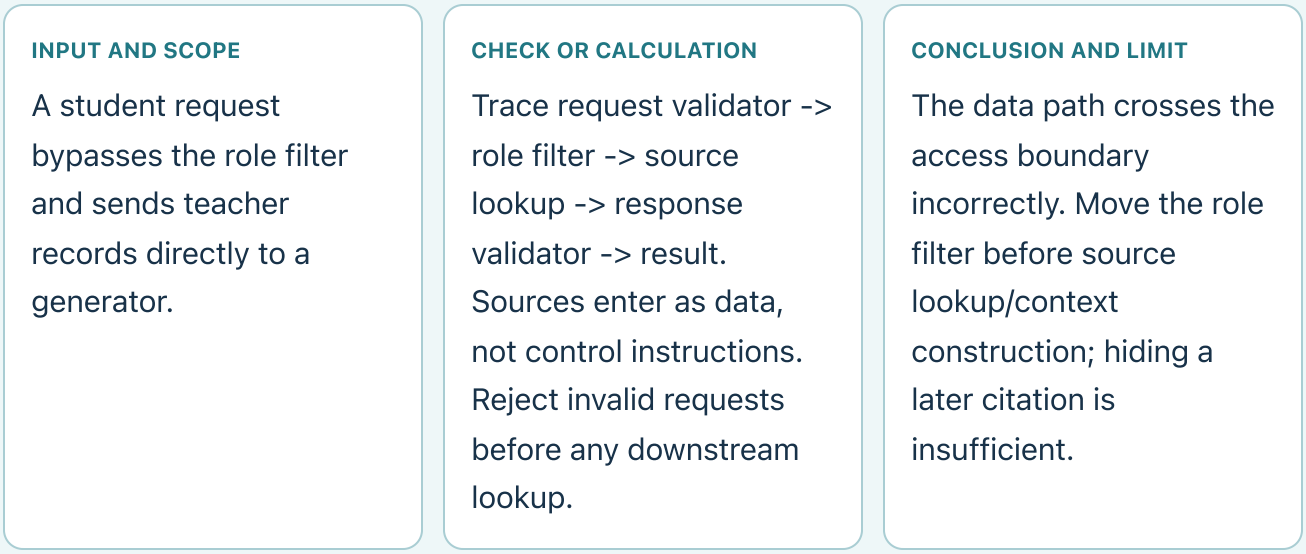
Local simulation fixture	Supplied value
Source S	student-visible, time 16:00
Source T	teacher-only, time 18:00

Local simulation fixture	Supplied value
Trusted caller role for main probes	student, provided separately by the test harness
Request shape	Exactly one field: source_id, a nonempty string
Valid response shape	source_id and time must exactly match the accessible source
No permitted source	review, without drafting a guessed time
Scope	Local deterministic simulation; no authentication, model call or deployed service

Require the access filter before source body lookup and context construction. A citation removed after generation does not repair earlier exposure. In the toy simulation role comes from the test harness rather than authentication; do not claim a deployed security check. Check denied T has no source_lookup/draft events, missing X has no draft, and invalid request shape stops at request_validator.

For the generic future-case question, a plan alone supplies no execution evidence; collect a fresh trace against fixed component contracts.

Architecture and data flow — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Qualification: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A student request bypasses the role filter and sends teacher records directly to a generator.

The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Normal and stopped trace expectations

Complete local probe	Expected stages and outcome
Student requests S	Validate → filter → lookup → draft → output check → answered S/16:00
Student requests teacher-only T	Validate → filter → denied; no source body lookup or drafting
Student requests missing X	Validate → filter → lookup → review; no drafting
S draft incorrectly says 19:00	Output check returns review, without publishing mismatched time
Request contains a caller-written role=teacher field	Invalid request shape; caller cannot promote their trusted role
Diagram claims access filtering but trace omits it	Implementation diverges from plan; preserve the failing trace and repair the route
A source text requests new permissions	Treat it as data, not an authorized control instruction; this simple simulator is not a complete attack defense

Separate library givens and scope

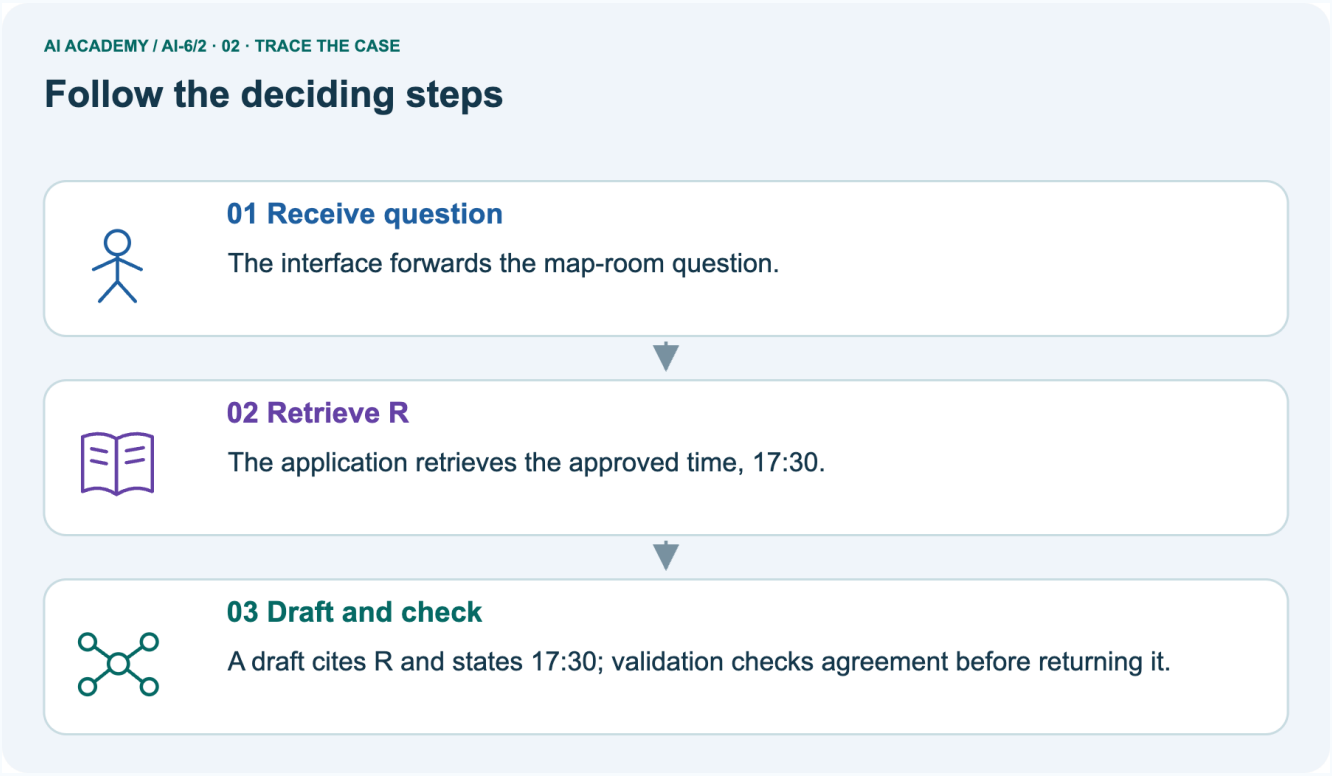
Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.

Approved schedule R says 17:30. The model may phrase facts but cannot replace them.

The main scenario diagnoses a student path that bypassed access filtering; restricted teacher records must not reach drafting context. The picture is a separate library reply: approved schedule R says the map room closes at 17:30. Retrieval supplies that fact, drafting supplies wording, and validation checks agreement. If retrieval is disconnected, return review rather than invent a time.

The picture’s R/17:30 is separate from the simulation’s S/16:00 and T/18:00. None of these fictional records verifies a current real closing time.

Library trace and disconnected-retrieval key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Receive question	The interface forwards the map-room question.
2. Retrieve R	The application retrieves the approved time, 17:30.
3. Draft and check	A draft cites R and states 17:30; validation checks agreement before returning it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An architecture assigns responsibilities and carries data between **component**s. Retrieval provides **evidence**; drafting supplies wording.

A drawing needs comparison with actual traces; it does not prove the implementation follows the arrows.

Change: Disconnect retrieval from the application.

Prediction: Return review status rather than inventing a time.

Reason: The supported-answer path lacks its required **evidence input**.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Component	A unit with a declared responsibility
Data flow	The path information takes through components
Trust boundary	A separation between differently authorised actors or data

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

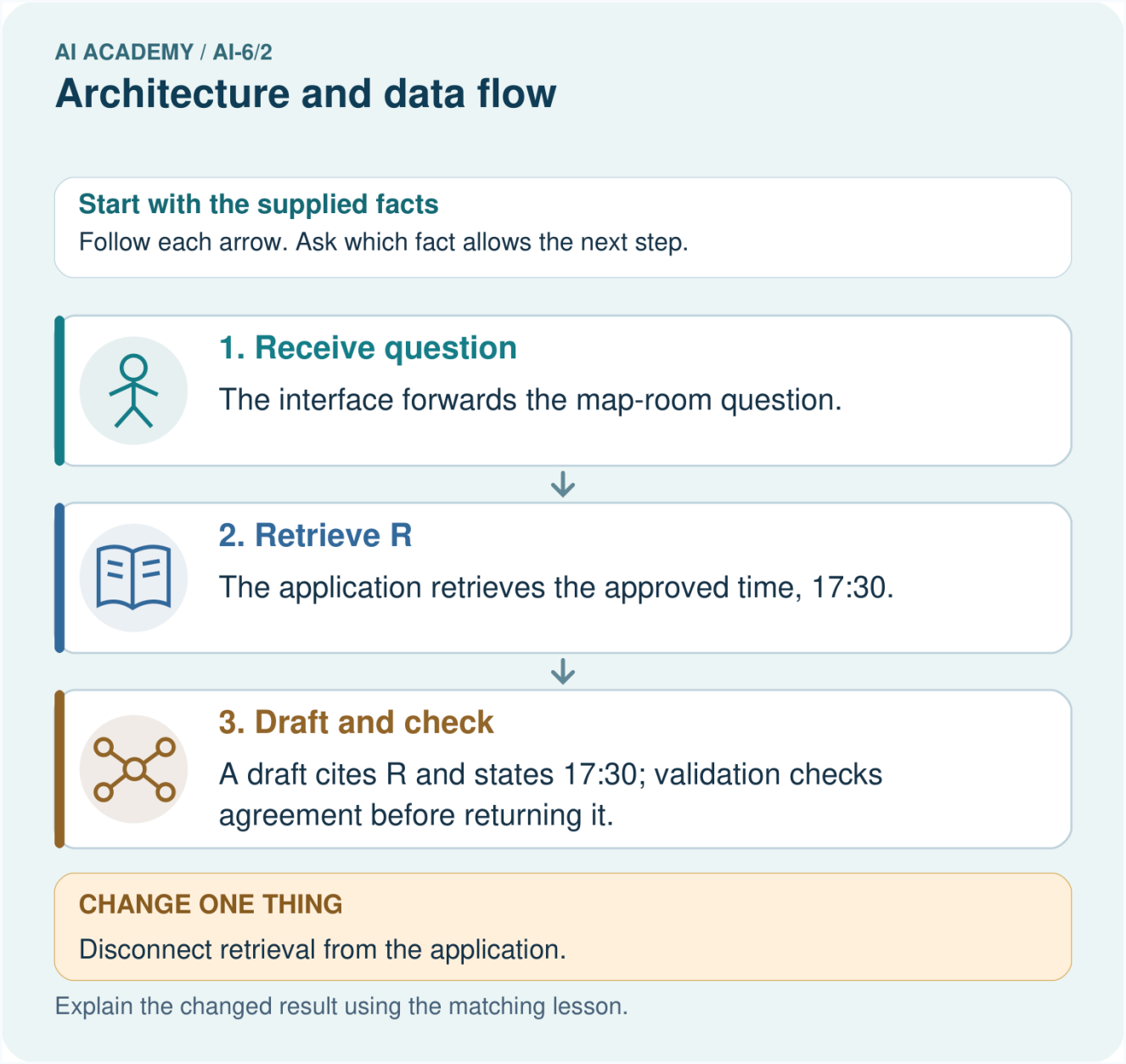
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Architecture and data flow

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A student request bypasses the role filter and sends teacher records directly to a generator. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Architecture and **data flow**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Label what moves along the library request, retrieval and drafting arrows. Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?

The interface forwards the map-room question. The application retrieves the approved time, 17:30. A draft cites R and states 17:30; validation checks agreement before returning it.

Changed case: Return review status rather than inventing a time. The supported-answer path lacks its required **evidence input**.

Changed-case reasoning: Return review status rather than inventing a time. The supported-answer path lacks its required **evidence input**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main boundary repair: filter before source body/context; removing citation later is insufficient. Local denied T stops before lookup/draft; missing X reviews before draft; mismatched S time reviews at output validation; client-added role field invalid at input validation. Library R17:30 is source-derived; disconnected retrieval reviews. Independent malformed request should stop at request validation before lookup.

Require stage order and explicit evidence limits, not merely a final refusal.

Diagnose the current misconceptions

The worked result establishes every future case: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: **Input** is malformed, yet lookup runs and then a validator rejects **output**. Locate the first stage that should stop.

Teacher explanation: Request validation should reject before lookup; **output** validation cannot undo unnecessary or unauthorised earlier reads.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Architecture assigns responsibilities to **component**s and shows information crossing trust boundaries. A diagram should include failure paths.

Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.

W2 Trace the worked case

A student request bypasses the role filter and sends teacher records directly to a generator. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

W3 Audit the limit

A report omits this qualification: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Qualification: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Input is malformed, yet lookup runs and then a validator rejects **output**. Locate the first stage that should stop.

Teacher explanation: Request validation should reject before lookup; **output** validation cannot undo unnecessary or unauthorised earlier reads.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Request validation should reject before lookup; **output** validation cannot undo unnecessary or unauthorised earlier reads. Require intermediate reasoning and the stated scope.

Fresh retry: An unauthorised role reaches storage before a response filter. Expected reasoning: Enforce access before storage results enter the **process**ing path; preserve a denied outcome.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled architecture and **data flow evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of architecture and **data flow**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Typed contracts and schemas. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this **component** forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/3, ai-6/5, ai-6/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of architecture and **data flow**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A student request bypasses the role filter and sends teacher records directly to a generator. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The data path crosses the access boundary incorrectly. Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Check every learner before discussion: Use the worked case for Architecture and **data flow**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Disconnect retrieval from the application. Require a **prediction** and a reason before comparing. Expected result: Return review status rather than inventing a time.

Reason: The supported-answer path lacks its required **evidence input**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

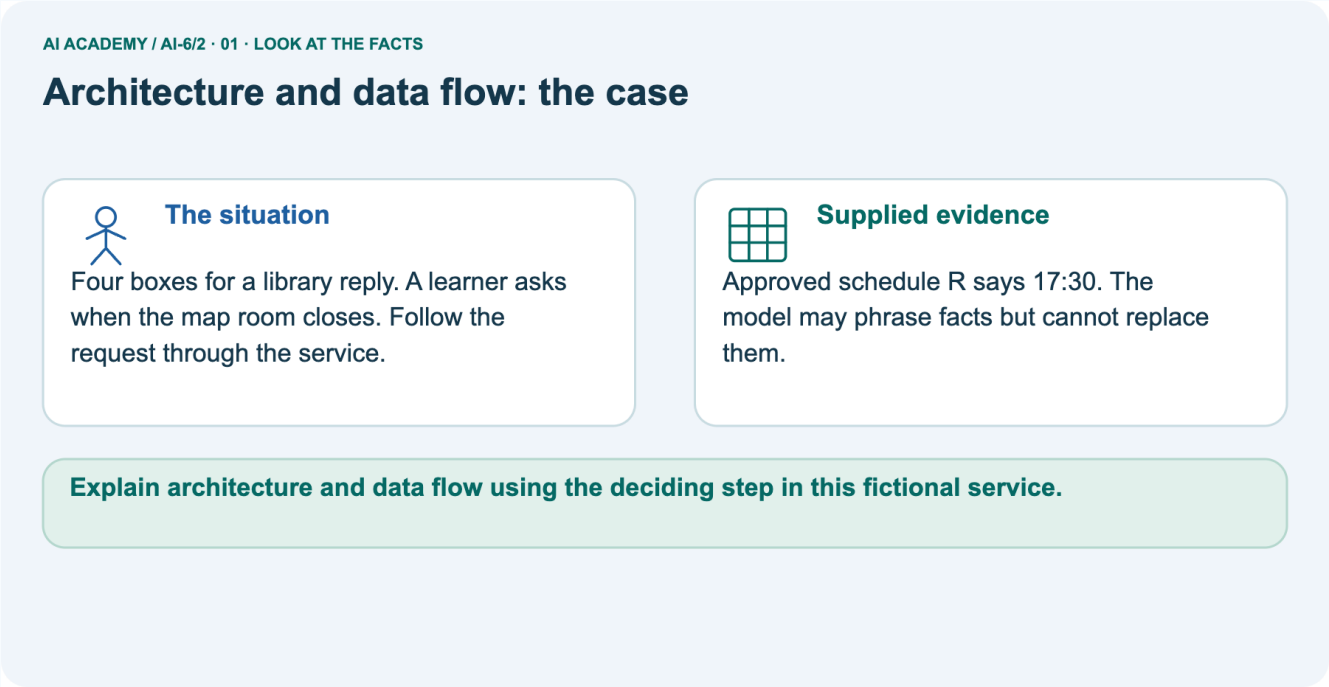
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Architecture and data flow: the case



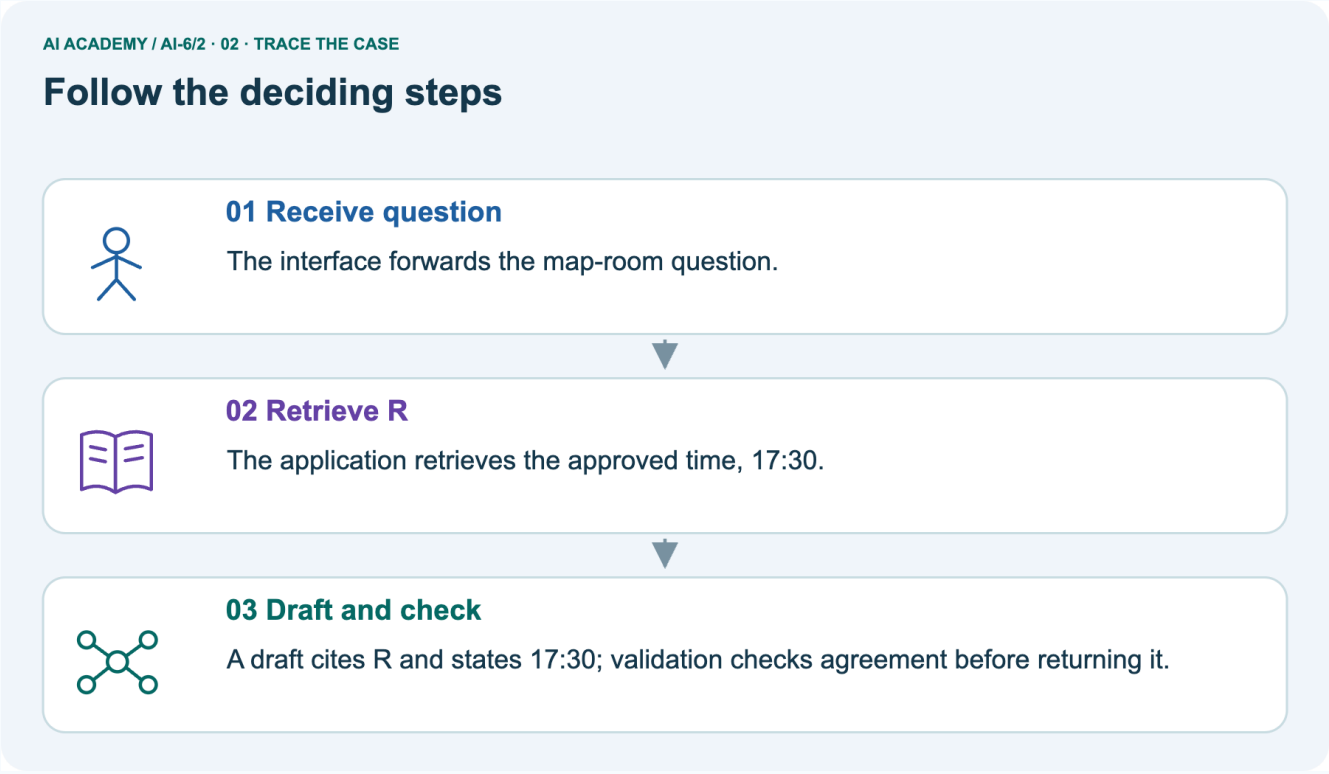
Follow the labelled picture with the matching explanation. Open the vector diagram

Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service. Supplied facts: Approved schedule R says 17:30. The model may phrase facts but cannot replace them.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



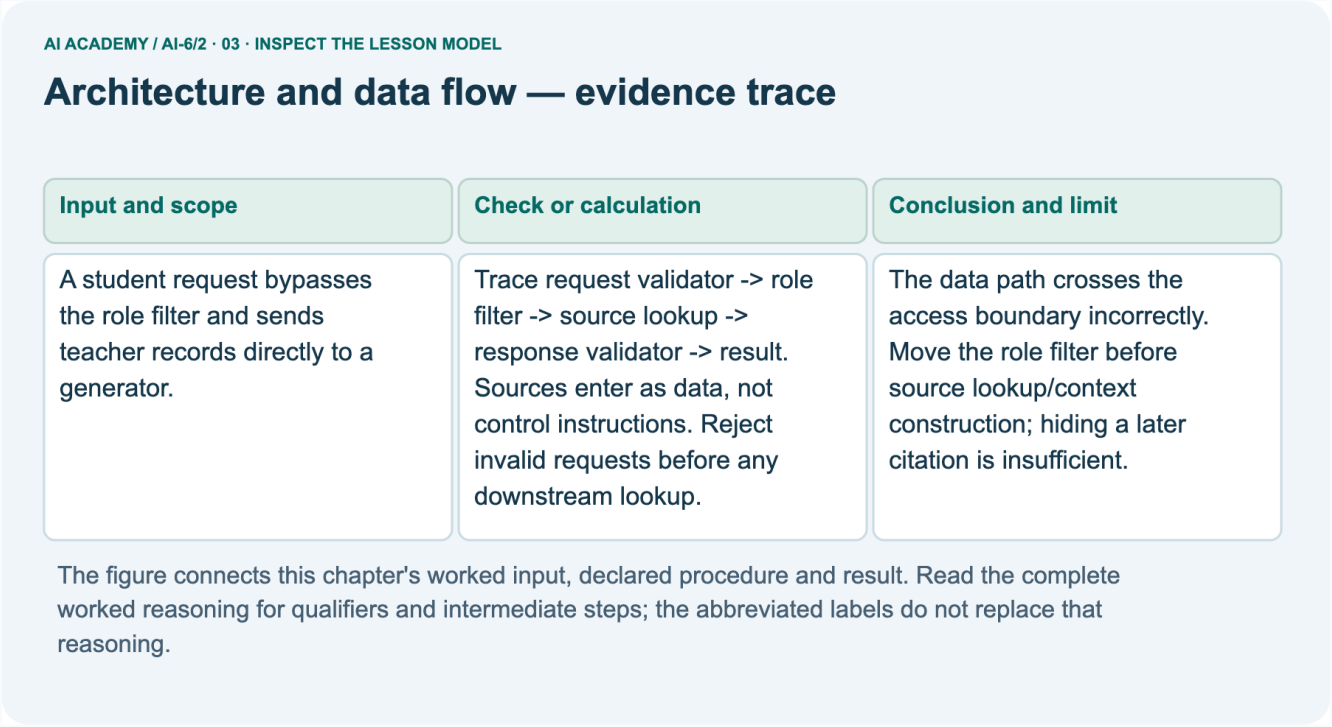
Follow the labelled picture with the matching explanation. Open the vector diagram

Receive question: The interface forwards the map-room question. → Retrieve R: The application retrieves the approved time, 17:30. → Draft and check: A draft cites R and states 17:30; validation checks agreement before returning it.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Architecture and data flow — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Architecture and **data flow** — **evidence** trace.

A student request bypasses the role filter and sends teacher records directly to a generator. | Trace request validator -> role filter -> source lookup -> response validator -> result.

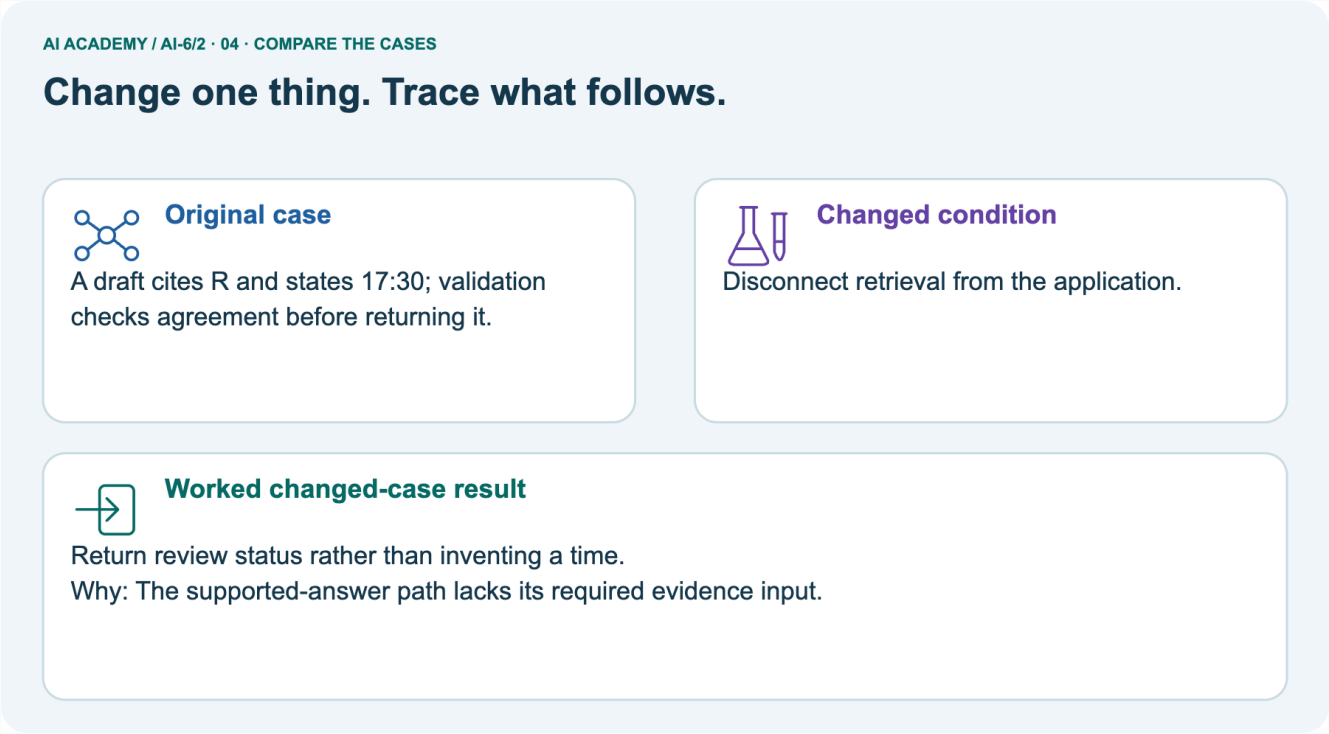
Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup. | The data path crosses the access boundary incorrectly.

Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A draft cites R and states 17:30; validation checks agreement before returning it. Changed condition: Disconnect retrieval from the application.

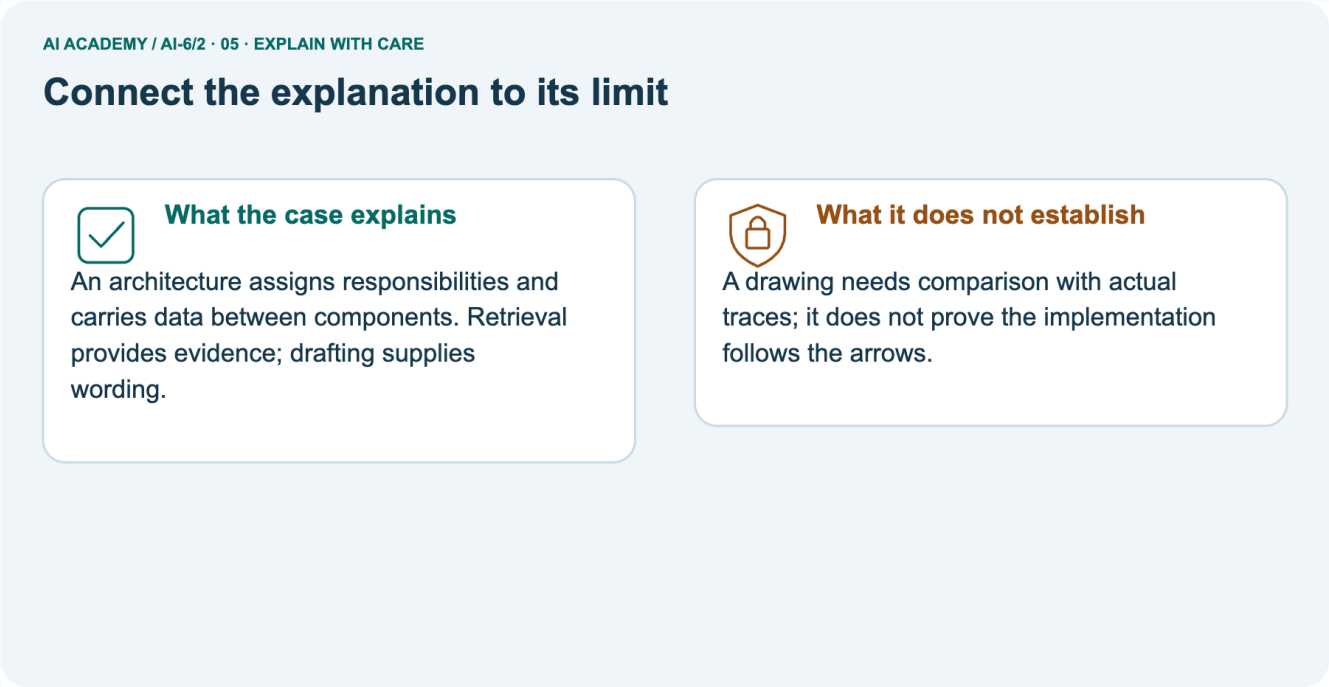
Expected result: Return review status rather than inventing a time. Reason: The supported-answer path lacks its required **evidence input**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The supported-answer path lacks its required **evidence input**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An architecture assigns responsibilities and carries data between **component**s. Retrieval provides **evidence**; drafting supplies wording.

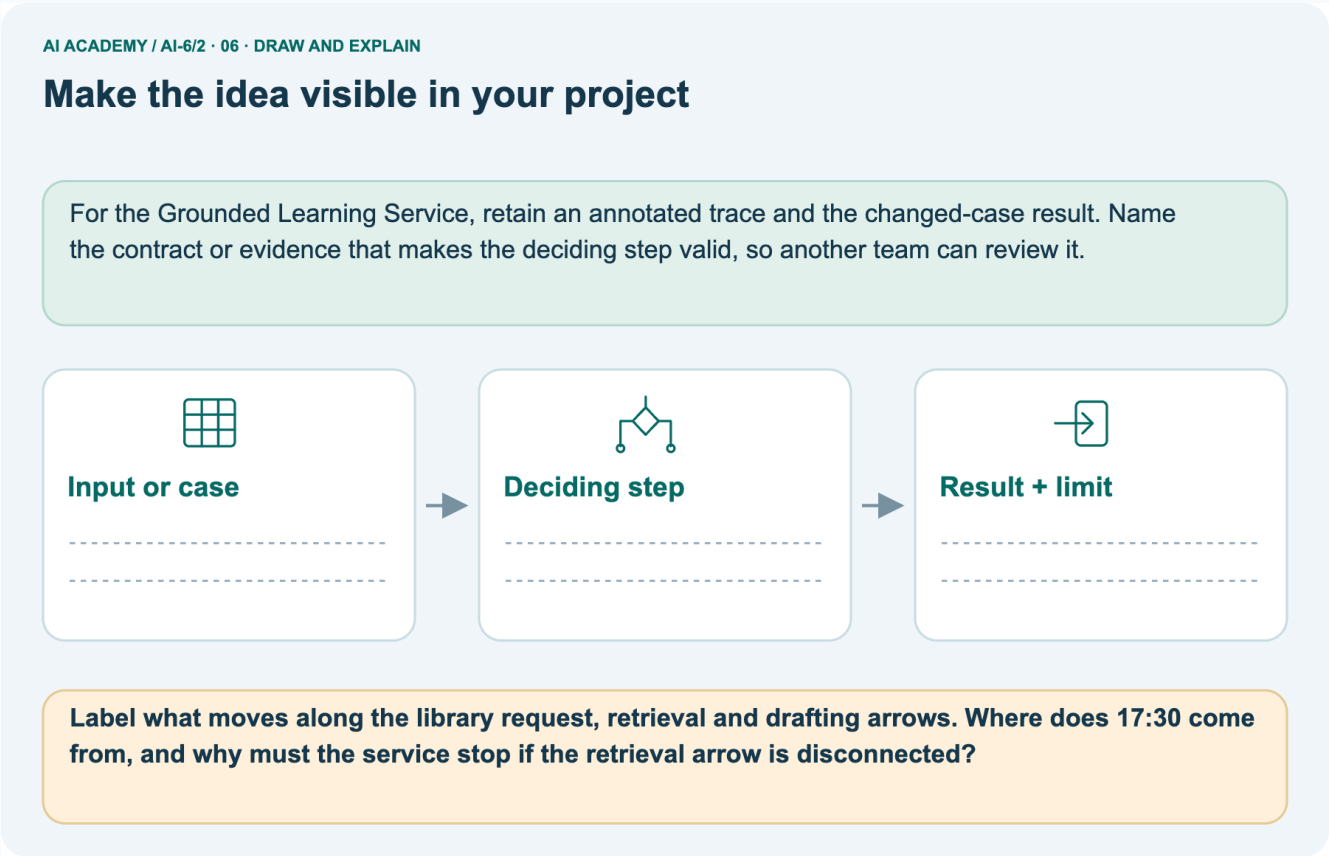
Limit: A drawing needs comparison with actual traces; it does not prove the implementation follows the arrows.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A drawing needs comparison with actual traces; it does not prove the implementation follows the arrows.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The interface forwards the map-room question. The application retrieves the approved time, 17:30. A draft cites R and states 17:30; validation checks agreement before returning it.

Changed case: Return review status rather than inventing a time. The supported-answer path lacks its required **evidence input**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Architecture and data flow: the case

02. Follow the deciding steps

03. Architecture and data flow — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.